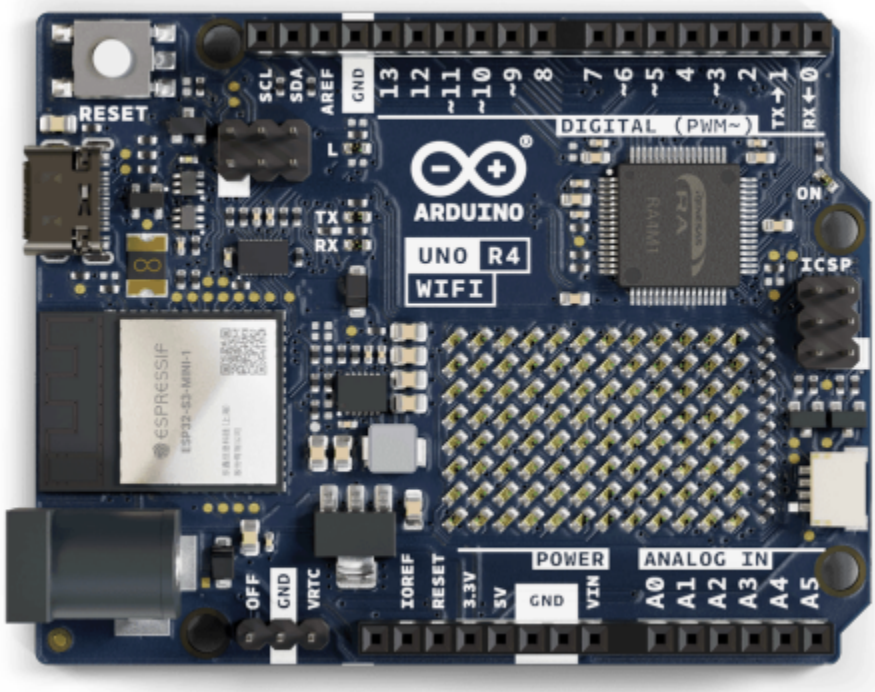
ArduiUnoR4

Link :
https://docs.arduino.cc/resources/datasheets/ABX00087-datasheet.pdf



- **Peripherals**
  - Capacitive Touch Sensing Unit (CTSU)
  - USB 2.0 Full-Speed Module (USBFS)
  - 14-bit ADC
  - Up to 12-bit DAC
  - Operational Amplifier (OPAMP)

- **Communication**
  - 1x UART (pin D0, D1)
  - 1x SPI (pin D10-D13, ICSP header)
  - 2x I2C (pin A4, A5, SDA, SCL), (Qwiic connector)
  - 1x CAN (pin D4, D5, external transceiver is required)

- **Wi-Fi®**
  - Wi-Fi® support with 802.11 b/g/n standard (Wi-Fi® 4)
  - Bit rate at up to 150 Mbps
  - 2.4 GHz band

- **Bluetooth®**
  - Bluetooth® 5

2 Recommended Operating Conditions

Symbol	Description	Min	Typ	Max	Unit
V_{IN}	Input voltage from VIN pad / DC Jack	6	7.0	24	V
V_{USB}	Input voltage from USB connector	4.8	5.0	5.5	V
T_{OP}	Operating Temperature	-40	25	85	°C

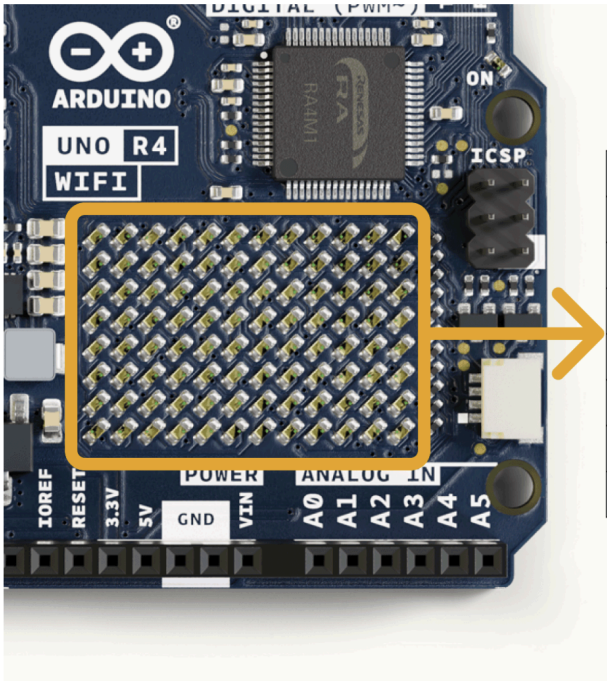
Note: V_{DD} controls the logic level and is connected to the 5V power rail. V_{AREF} is for the analog logic.

Entry-level projects: If this is your first project within coding and electronics, the UNO R4 WiFi is a good fit. It is easy to get started with, and it has a lot of online documentation.

Easy IoT applications: build projects without writing any networking code in the Arduino Cloud. Monitor your board, connect it with other boards and services, and develop cool IoT projects.

LED Matrix: the 12x8 LED matrix on the board can be used for showing animations, text scrolling, create mini-games and much more, being the perfect feature to give your project more personality.

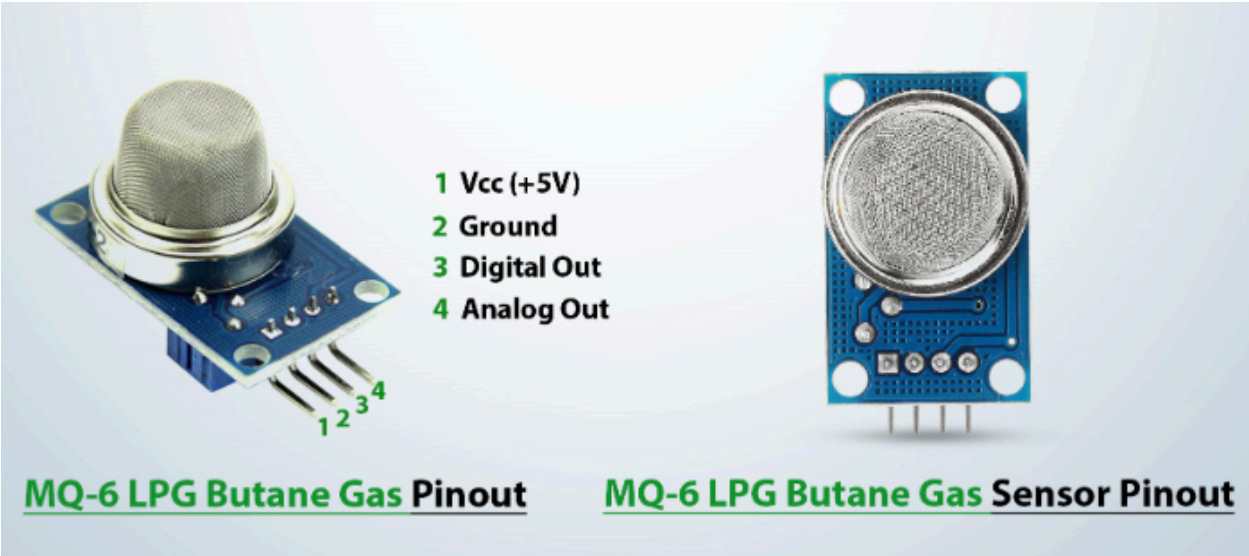
These LEDs can be accessed as an array, using a specific library. See the mapping below:



1	2	3	4	5	6	7	8	9	10	11	12
13	14	15	16	17	18	19	20	21	22	23	24
25	26	27	28	29	30	31	32	33	34	35	36
37	38	39	40	41	42	43	44	45	46	47	48
49	50	51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70	71	72
73	74	75	76	77	78	79	80	81	82	83	84
85	86	87	88	89	90	91	92	93	94	95	96

MQ6

Link: http://www.handsontec.com/dataspecs/sensor/MQ6-LPG%20Sensor%20Module.pdf



Brief Description:

- Power Supply: 5VDC.
- Model No: MQ-6.
- Heater voltage: 5±0.2V.
- Heating Resistance: 33Ω±3% (Room temperature).
- Heating Power: ≤ 750mW.
- Sensing Resistance: 10KΩ~ 60KΩ (1000ppm LPG).

Specification:

A. Standard work condition

Symbol	Parameter name	Technical condition	Remarks
Vc	Circuit voltage	5V±0.1	AC OR DC
V_H	Heating voltage	5V±0.1	ACOR DC
P_L	Load resistance	20KΩ	
R_H	Heater resistance	33Ω ±5%	Room Tem
P_H	Heating consumption	less than 750mw	

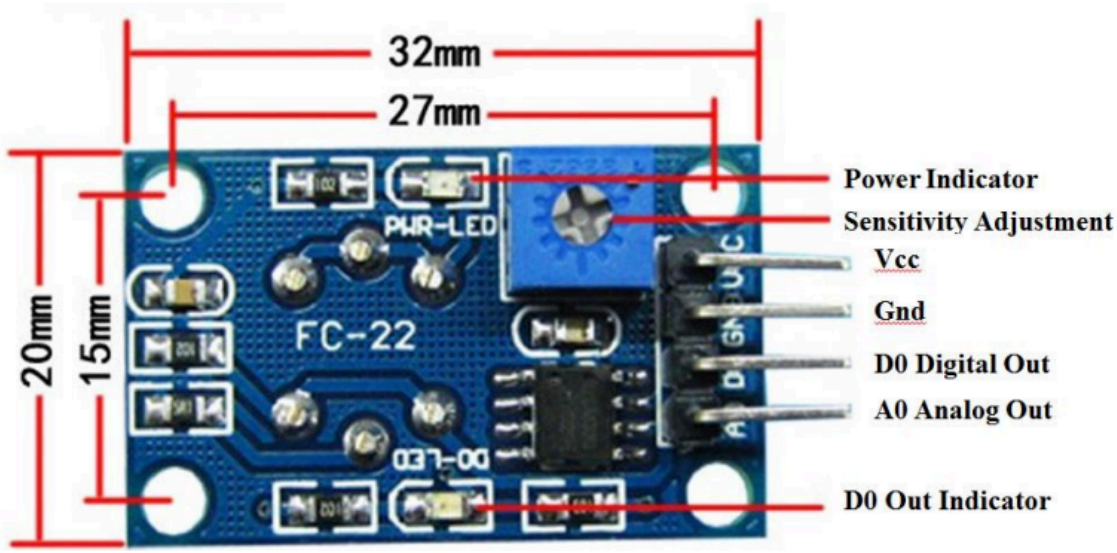
B. Environment condition

Symbol	Parameter name	Technical condition	Remarks
Tao	Using Tem	-10℃-50℃	
Tas	Storage Tem	-20℃-70℃	
R_H	Related humidity	less than 95%Rh	
O_2	Oxygen concentration	21%(standard condition)Oxygen concentration can affect sensitivity	minimum value is over 2%

C. Sensitivity characteristic

Symbol	Parameter name	Technical parameter	Remarks
Rs	Sensing Resistance	10K Ω - 60K Ω (1000ppm LPG)	Detecting concentration scope: 200-10000ppm LPG , iso-butane, propane, LNG
α (1000ppm/ 4000ppm LPG)	Concentration slope rate	≤0.6	
Standard detecting condition	Temp: 20℃±2℃ Humidity: 65%±5%	Vc:5V±0.1 Vh: 5V±0.1	
Preheat time	Over 24 hour		

Mechanical Dimension:



Working:

The voltage that the sensor output changes according to the smoke/gas level that exists in the atmosphere. The sensor outputs a voltage that is proportional to the concentration of smoke/gas.

The relationship between voltage and gas concentration is the following:

- The greater the gas concentration, the greater the output voltage
- The lower the gas concentration, the lower the output voltage

The output can be an analog signal (A0) that can be read with an analog input of the Arduino or a digital output (D0) that can be read with a digital input of the Arduino.